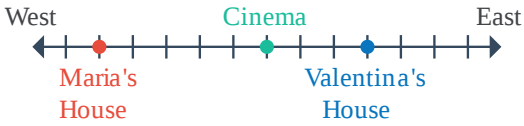
Number lines

1 Valentina and Maria want to meet at the cinema to watch a movie. Valentina lives 3 km west of the cinema, and Maria lives 5 km east of the cinema.

a In the number lines below, the distance between each tick is 1 km. Which of the following number lines show the location of Valentina and Maria in relation to the cinema?

A



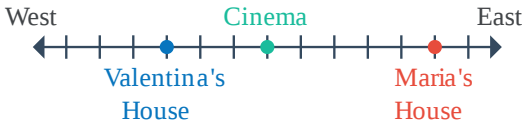
B



C



D



b Draw a new number plane that shows the location of Valentina and Maria using integers, where 0 is the cinema, east is the positive direction and west is the negative direction of the number plane.

c Calculate how far would Maria have to travel to get to Valentina's house.

2 At the start of the month, Laura owes $\$60$ to the bank. A week later, her employer deposits $\$130$ into her account.

a Sketch a number line that shows the change in Laura's account balance.

b Suppose that next month Laura now has $\$130$ in her account, and she spends $\$60$ on rent. Sketch a number line that shows the change in Laura's account balance for this month.

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5 The motion of a robot to the left or right along a straight line is determined by whether the robot is facing left or right, and whether it is stepping forwards or backwards.

Facing	Stepping	Moving
Right	Forward	Right
Right	Backward	Left
Left	Forward	Left
Left	Backward	Right

Use a number line where 1 unit represents 1 step and the positive direction is to the right to answer the following questions:

a For each of the following, find the integer that represents the final position:

i The robot starts at 0, facing to the right, and takes 5 steps forward.

ii The robot starts at 0, facing to the left, and takes 13 steps forward.

iii The robot starts at 0, facing to the left, and takes 11 steps backwards.

b Suppose the robot performs the following instructions in the given order:

- Start at -5, facing right.
- Take 7 steps forward.
- Take 21 steps backward.
- Turn to face left.
- Take 22 steps backward.

State the integer that represents its final position.

Integers

6 William arrives to a meeting 15 minutes early, and he leaves at the end, 75 minutes after he got there.

Find how long the meeting went for.

7 Maximilian is going to cook a pot roast. He turns the oven on to warm it up. He puts the pot roast in the oven 15 minutes later when it is at the right temperature. The pot roast is ready 120 minutes after Maximilian first turned the oven on.

Calculate how long the pot roast baked for.

8 Tara is waiting for the next flight to Los Angeles, which was scheduled to be in 36 minutes, but there is a 48 minute delay. She takes a nap, and wakes up 24 minutes later. Calculate how much longer Tara has to wait before the plane departs.

9 The planet Anarres experiences drastic changes in temperature over the course of the day. At sunrise the temperature is measured to be -250°C, and it warms to a temperature of -110°C by midday. Calculate the difference of temperature from sunrise to midday.

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10 A substance experiences rapid changes in temperature over the course of an experiment to test its electrical conductivity. At the start of the experiment the substance is measured to be 70°C, and by the end of the experiment its temperature is measured as -50°C.

Calculate the difference of temperature from the start to the end of the experiment.

11 A submarine is initially $45\ \text{m}$ below sea level. It then descends $10\ \text{m}$ straight down to its final position.

a If sea level is represented by the integer 0, and upwards is the positive direction, find the integer that represents the submarine's final position.

b Describe the final position of the submarine.

12 A bird soaring at a height of $25\ \text{m}$ above sea level sees a school of fish and dives down $32\ \text{m}$. Calculate how far below sea level the bird reaches.

13 When Maria woke up, the temperature was -4°C. During the day, it rose by 11°C but during the evening up until she went to bed, it also dropped by 5°C. Calculate the temperature when Maria went to bed.

14 A bus departs the depot with 8 passengers on board. At the first stop, 4 people got off and 6 people got on. At the second stop, a further 3 people got off, and 7 more got on. Calculate the number of people are on the bus when it leaves the second stop.

15 Luigi enters an elevator at the 7th floor (the ground floor being floor 0). The elevator goes down 3 floors, then up 9 floors and finally it goes down 2 floors, where Luigi gets out.

a State which floor does Luigi end up.

b When Luigi gets off the elevator, how many floors from his starting point is he?

16 At the start of the week, Quiana had $\$131$ in her bank account. On Monday, she withdrew $\$43$. On Wednesday, she deposited $\$11$, and on Thursday, she deposited a further $\$28$. Finally, she withdrew $\$25$ on Friday.

Find Quiana's final bank balance at the end of the week.

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17 Lisa, Xavier and Frank are in the final round of a math quiz. The quiz consists of 6 questions. The scoring system for each round is as follows:

- 5 points for every correct answer.
- A loss of 3 points for every incorrect answer.
- 0 point for every unanswered question.

The table below summarizes the results of the final round:

Question	1	2	3	4	5	6
Lisa	Correct	Incorrect	No answer	Correct	Incorrect	Correct
Xavier	No answer	No answer	Correct	Correct	Incorrect	Correct
Frank	Correct	Correct	Incorrect	Incorrect	Incorrect	Correct

a Find the final score of:

i Lisa **ii** Xavier **iii** Frank

b Who won the quiz?

18 Starting from his house, Jerry walked $1\ \mathrm{m/s}$ due south for 30 minutes, then $2\ \mathrm{m/s}$ due north for 15 minutes and then, then due south again $2\ \mathrm{m/s}$ for 45 minutes. Find Jerry's final distance from his house.

19 Jake had $\$80$ at the beginning of the day. He then bought bread for $\$5$ and milk for $\$3$. Then, he went to a game shop and bought a game for $\$50$. On his way home he found $\$10$. How much money does Jake have now?

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